

Telecom Customer Churn Analysis

Exploratory Data Analysis and Retention Insights

1. Executive Summary

This report summarizes an exploratory analysis of a telecom customer churn dataset containing **7,043 customers and 21 variables**. The objective was to identify customer segments and service characteristics associated with higher churn and translate those patterns into practical retention priorities.

The analysis found an overall churn rate of **26.54%**: 1,869 customers churned and 5,174 were retained. The clearest patterns were concentrated among month-to-month customers, fiber optic subscribers, customers with shorter tenure, and customers without Online Security or Technical Support.

These findings are associations observed in the dataset. They do not establish causation. They are best used to guide customer segmentation, deeper operational investigation, and future predictive or experimental work.

Key findings

- **Month-to-month contracts:** approximately 42.7% churn, compared with 11.3% for one-year and 2.8% for two-year contracts.
- **Fiber optic service:** approximately 41.9% churn, compared with 19.0% for DSL and 7.4% for customers with no internet service.
- **Online Security:** customers without the service show approximately 41.8% churn, compared with 14.6% among subscribers.
- **Technical Support:** customers without the service show approximately 41.6% churn, compared with 15.2% among subscribers.
- **Tenure:** shorter-tenure customers show greater churn vulnerability, making the first year an important retention window.

2. Project Objective and Scope

The project was designed as a Python-based exploratory data analysis. It focuses on measuring churn, comparing churn rates across customer segments, and identifying areas that merit retention attention.

- Measure the overall customer churn baseline.
- Compare churn across contract types and internet service categories.
- Evaluate churn patterns across telecom service attributes.
- Segment customers by tenure to examine lifecycle-related churn patterns.
- Translate observed patterns into business recommendations without overstating causal conclusions.

3. Data Quality and Preparation

Before calculating churn rates, the dataset was inspected for structure, missing values, duplicate records, and data-type issues. The notebook uses a relative dataset path so the analysis can be rerun after cloning the repository.

Dataset quality checks

- Dataset shape confirmed as **7,043 rows × 21 columns**.
- Complete-row duplicate check returned **0 duplicates**.
- Duplicate customerID check returned **0 duplicate identifiers**.
- Standard null checks initially reported no conventional null values.

TotalCharges handling

The TotalCharges field was initially stored as text because blank-string values were present. It was converted using `pd.to_numeric(..., errors='coerce')`, which exposed **11 missing numeric values**. Inspection showed that these 11 records had tenure = 0.

For this exploratory analysis, the 11 missing TotalCharges values were imputed as 0. This is consistent with the observed zero-tenure records and preserves those customers in the analysis. In a production environment, the assumption should be validated against billing-system definitions.

Categorical standardization

The SeniorCitizen indicator was mapped from 1/0 to Yes/No for clearer interpretation and consistency with other categorical fields.

4. Analytical Findings

4.1 Overall churn

Of 7,043 customers, **1,869 churned** and **5,174 were retained**. The resulting churn rate is **26.54%**. Approximately one in four customers in the dataset left the provider.

Business interpretation: the churn baseline is large enough to justify targeted retention analysis. Segment-level results help identify where retention efforts may be more relevant.

4.2 Contract type

Contract type	Churn rate
Month-to-month	42.7%
One year	11.3%
Two year	2.8%

Month-to-month customers have the highest churn rate by a wide margin. Longer contract duration is associated with lower churn in this dataset.

Retention priority: test contract-upgrade incentives and loyalty offers for month-to-month customers. Any campaign impact should be measured before broader rollout.

4.3 Internet service

Internet service	Churn rate
Fiber optic	41.9%
DSL	19.0%
No internet service	7.4%

Fiber optic customers show the highest churn rate among internet-service groups. The analysis does not identify the cause of this difference.

Retention priority: investigate pricing, customer expectations, service experience, and support interactions for fiber optic customers using additional operational data.

4.4 Online Security and Technical Support

Service attribute	No service	Subscribed
Online Security	41.8%	14.6%
Technical Support	41.6%	15.2%

Customers without Online Security or Technical Support have materially higher churn rates than customers subscribed to those services. These variables are useful segmentation signals, but the analysis does not show that adding either service will directly prevent churn.

Retention priority: use service-adoption gaps as part of high-risk segmentation and test relevant support or bundle offers with controlled measurement.

4.5 Customer tenure

Tenure was grouped into 0–12, 13–24, 25–48, and 49–72 month segments. The analysis shows greater churn vulnerability among shorter-tenure customers, while longer-tenure groups are more strongly associated with retention.

Retention priority: treat the first year as a key lifecycle window for onboarding, proactive support, and satisfaction monitoring.

4.6 Additional service analysis

The notebook also compares churn across PhoneService, MultipleLines, OnlineBackup, DeviceProtection, StreamingTV, and StreamingMovies. These service variables provide additional segmentation context for future multivariable or predictive analysis.

5. Business Recommendations

- 1. Prioritize month-to-month customers.** Build a retention segment around flexible-contract customers and test annual-plan incentives, loyalty benefits, or targeted offers.
- 2. Investigate fiber optic churn.** Combine churn data with billing, complaints, support, and service-quality data to identify why this group shows elevated churn.
- 3. Strengthen early-tenure engagement.** Review onboarding and proactive support during the first 12 months.
- 4. Use support and security gaps as segmentation signals.** Identify high-risk customers without Online Security or Technical Support and test relevant service education or bundle offers.
- 5. Measure retention actions.** Track churn, offer acceptance, retention uplift, and customer value rather than assuming an EDA association will produce a causal effect.

6. Limitations

This project is exploratory. The analysis identifies associations between customer attributes and churn but does not establish causation. Operational factors that could explain churn—such as complaint history, outage frequency, customer satisfaction, competitor offers, and detailed pricing changes—are not available in the dataset.

The findings should therefore be used to prioritize deeper investigation and retention experiments. A future predictive model would require careful feature preparation, train/test separation, class-aware evaluation, and monitoring for changes in customer behavior over time.

7. Next Steps

- Create explicit tenure-based retention segments for operational reporting.
- Develop a Power BI dashboard for churn KPI and segment monitoring.
- Reproduce key churn aggregations in SQL as a separate analytical workflow.
- Engineer candidate features and build a transparent baseline classification model.
- Evaluate predictive performance using precision, recall, F1-score, and ROC-AUC.
- Use churn probabilities to support high-risk customer prioritization, subject to business validation.

8. Conclusion

The analysis identifies a clear set of churn segmentation patterns. Month-to-month customers, fiber optic subscribers, shorter-tenure customers, and customers without Online Security or Technical Support show higher churn in the dataset.

The strongest immediate analytical opportunity is to focus retention investigation on short-tenure, month-to-month customers and to examine the elevated churn observed among fiber optic subscribers. The results provide a practical foundation for targeted retention analysis, operational validation, and future predictive modeling.

Prepared by

Srinadhu Krishna Prabath

Data Analyst | Python | SQL | Power BI | Excel